SatyaScore PRD v2.4

- PRODUCT REQUIREMENTS DOCUMENT
  - SatyaScore: Multilingual Hallucination Detection & Scoring Framework
  - Table of Contents
  - 1. Executive Summary
  - 2. Problem Statement & Motivation
  - 3. Positioning Against Prior Art
  - 4. Project Overview
  - 5. Goals & Success Metrics
  - 6. Stakeholders & Team Roles
  - 7. System Architecture
  - 8. Core Functional Requirements
  - 9. Non-Functional Requirements
  - 10. Technology Stack (with Concrete Hardware Specs)
  - 11. Dataset Strategy
  - 12. Development Roadmap (4 Months / ~17 Weeks)
  - 13. Research Paper Strategy
  - 14. Patent Strategy
  - 15. Risk Assessment & Mitigation
  - 16. Ethical & Responsible-AI Considerations
  - 17. Appendix: Key Reference Papers (Updated — Now 27 Papers Across 5 Categories)

PRODUCT REQUIREMENTS DOCUMENT

SatyaScore: Multilingual Hallucination Detection & Scoring Framework

Real-Time LLM Trustworthiness Verification for English, Hindi (core) and Marathi (tiered stretch goal)

Final Year Capstone Project | B.Tech Artificial Intelligence & Data Science PRMIT&R Badnera-Amravati | Guide: Dr. S. R. Gupta Version 2.4 | Revised July 2026 (supersedes v2.3) — Project renamed to SatyaScore Team Size: 4 Members | Timeline: 4 Months (~17 weeks) Targets: IEEE Access / ACL-adjacent Publication + Indian Provisional Patent

> **What changed in v2.0:** After a full literature survey (22+ papers across foundational hallucination detection, the Mu-SHROOM shared task ecosystem, SHROOM-CAP, cross-lingual embeddings, and fact-verification/RAG research), the project's novelty claim was sharpened. v1.0 claimed to be "the first" multilingual hallucination system for Indian languages. That claim doesn't hold — Mu-SHROOM (SemEval-2025 Task 3) already benchmarks Hindi at scale across 14 languages, and SHROOM-CAP (2025) extends this to 9 languages including 5 Indic ones. v2.0 repositioned novelty around Marathi coverage, knowledge-graph-grounded verification, and graded claim-level scoring combined together.
>
> **What changed in v2.1:** Given the 6-month timeline, building and annotating a full 3,000+ claim Marathi dataset from scratch alongside everything else the project requires was judged infeasible. **Marathi has been moved from a core deliverable to a future-work / stretch goal.** The committed core scope is now **English + Hindi only**. The novelty claim shifts accordingly: it now rests primarily on **knowledge-graph fusion (semantic + NLI + KG combined into one graded score)** and **claim-level granularity**, rather than on being the only system to cover Marathi. This is still a genuinely defensible contribution — no reviewed system (Mu-SHROOM, SHROOM-CAP, or any foundational work) combines all three signals at claim level for any Indian language, Hindi included. Marathi is retained in the document as an explicitly flagged extension path, since the architecture is designed to generalize to it later with minimal rework.
>
> **What changed in v2.2:** A deeper search surfaced closer prior art than previously known: **MultiHal** (Lavrinovics et al., 2025), **FactKG** (Kim et al., 2023), and **KGR** (Guan et al., 2024) all touch knowledge-graph-grounded hallucination verification. None cover Hindi, and none fuse semantic+NLI+KG into one deployable claim-level score with a real-time API — so the core novelty claim survives — but it must now be stated more precisely and must explicitly address these three works rather than claiming KG-grounding as untouched territory. A new reference category (Category E) was added in Section 17 to reflect this.
>
> **What changed in v2.3 (this version, after strict internal review):** Confirmed timeline is a full 4 months (~17

> weeks), not under 4. **Marathi is reinstated as a lighter, tiered "stretch core" goal** — a smaller dataset (800–1,200 claims) explicitly ranked below the English/Hindi deliverables and the ablation study in priority, not full parity. This version also fixes several internal inconsistencies caught in a strict critical review: (a) corrected Month-based patent/paper deadlines that didn't match the compressed weekly roadmap, (b) added MultiHal/FactKG/KGR visibility to the Executive Summary instead of burying it in Section 3, (c) added a KG-only baseline derived from FactKG's approach so the closest prior art is actually benchmarked against, not just cited, (d) rebalanced the roadmap so annotation runs in parallel with pipeline development instead of being crammed into one phase, (e) added a risk entry for the ablation study itself showing a null result, (f) specified the actual weight-learning mechanism (softmax-constrained, not plain logistic regression), (g) re-scoped the FAISS index size question for a trilingual (not 5-language) corpus, (h) rebalanced team roles to un-overload Member 4, (i) added a compute cost acknowledgment.
>
> **What changed in v2.4 (this version):** Project renamed from "MultiHallu-Score" to **SatyaScore** throughout the document. No technical, scope, or timeline changes — this is a naming update only. The underlying algorithm retains its name, CMHSA (Composite Multilingual Hallucination Scoring Algorithm).

Table of Contents

1. Executive Summary
2. Problem Statement & Motivation
3. Positioning Against Prior Art
4. Project Overview
5. Goals & Success Metrics
6. Stakeholders & Team Roles
7. System Architecture
8. Core Functional Requirements
9. Non-Functional Requirements
10. Technology Stack (with Concrete Hardware Specs)
11. Dataset Strategy
12. Development Roadmap (4 Months / ~17 Weeks)
13. Research Paper Strategy
14. Patent Strategy
15. Risk Assessment & Mitigation (incl. 15.1 Marathi Tiered Stretch-Goal Plan)
16. Ethical & Responsible-AI Considerations
17. Appendix: Key Reference Papers (Updated)

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1. Executive Summary

Large Language Models (LLMs) such as GPT-4, Gemini, and Llama are increasingly deployed in critical domains including healthcare, law, education, and governance. A persistent and dangerous failure mode of these models is hallucination — the generation of plausible but factually incorrect or fabricated content, delivered with high confidence.

This project proposes the design, development, and evaluation of **SatyaScore**, a claim-level Hallucination Detection and Scoring Framework with **English and Hindi as core, committed languages**, and **Marathi as a tiered stretch goal** (a smaller ~800–1,200 claim dataset, explicitly ranked below the core deliverables — see Section 15). The system decomposes LLM-generated text into atomic factual claims, verifies each claim against a structured knowledge base, and produces a graded (continuous) hallucination confidence score at the individual-claim level — rather than a binary label or an approximate error span.

The core technical contribution is the **Composite Multilingual Hallucination Scoring Algorithm (CMHSA)**, which fuses three independent verification signals — cross-lingual semantic similarity, natural language inference (NLI) entailment, and knowledge-graph consistency verification (Wikidata) — into a single interpretable score. This is, to the best of current literature, the first system to combine all three signal types at the claim level for Hindi or English, a combination absent from Mu-SHROOM (similarity/entailment/voting only, no KG), SHROOM-CAP (binary labels, no KG evidence, no graded score), and more recent knowledge-graph-grounded work — **MultiHal, FactKG, and KGR** — none of which fuse all three signals into one deployable, claim-level score for Hindi (see Section 3 and Section 17 Category E for full differentiation). This dual-purpose project targets both a Provisional Indian Patent application and a research paper submission to a peer-reviewed NLP venue (IEEE Access, or an ACL-affiliated workshop track appropriate for a student capstone).

2. Problem Statement & Motivation

2.1 The Hallucination Problem

Hallucination in LLMs refers to the generation of confident, syntactically fluent, yet factually incorrect or entirely fabricated content. Reported hallucination rates vary widely (roughly 15%–40% of factual queries) depending on domain, model, and language — and rates are generally worse for lower-resource languages such as Hindi and Marathi, where training data is comparatively sparse.

Current, still-unresolved challenges in hallucination detection include: - No production-ready, real-time hallucination detector exists that is both **claim-level** and **knowledge-graph-grounded**, for English or Hindi. - Existing multilingual systems either mark approximate error locations (span-level, Mu-SHROOM) or apply a binary correct/incorrect label (SHROOM-CAP) — neither produces a graded, decision-support-ready confidence score. - No existing system combines semantic similarity, NLI entailment, **and** structured knowledge-graph verification into one fused score at the claim level, for any Indian language including Hindi. - Marathi — spoken by 80+ million people — has zero coverage in any major multilingual hallucination benchmark to date; this project addresses it at a smaller, tiered scale (Section 15) rather than full parity with Hindi/English, given the timeline.

2.2 Why This Matters

- India has 500M+ smartphone users consuming information in regional languages via LLM-powered apps.
- Government services, healthcare advisory, and legal information delivered via AI chatbots can cause serious harm if hallucinated content goes undetected.
- No Indian patent currently exists for a claim-level, knowledge-graph-grounded LLM hallucination scoring system for Hindi — a genuine IP opportunity remains open around the KG-fusion approach itself, independent of language coverage.

3. Positioning Against Prior Art

This section replaces the v1.0 “Research Gap” table, which is now outdated in light of the full literature survey.

Dimension	**Foundational Work (FEVER, FActScore, SelfCheckGPT, TruthfulQA, HaluEval, TRUE)**	**Mu-SHROOM (SemEval-2025)**	**SHROOM-CAP (2025)**	**SatyaScore (This Project)**
Language Coverage	English only	14 languages incl. Hindi	9 languages incl. 5 Indic	**English + Hindi** (Marathi: future work)
Detection Granularity	Whole response / single fact	Span-level (exact error location)	Whole document, binary	**Claim-level, graded score**
Verification Signals	One signal only (similarity OR entailment OR consistency)	Similarity, entailment, or LLM-ensemble voting	Not clearly evidenced	**Semantic + NLI + Knowledge Graph, fused**
Knowledge Graph Grounding	No	No	Not evident	**Yes (Wikidata SPARQL)**
Score Type	Binary or single-signal score	Span labels	Binary (yes/no)	**Continuous [0,1] + 3-tier verdict**
Domain	General/biographical	General AI-generated answers	Scientific publications only	**General-domain (healthcare, governance, education)**

Resulting novelty claim (defensible): SatyaScore is the first system to combine claim-level decomposition and three-signal fused scoring — including knowledge-graph verification — in a single general-domain framework for Hindi (or English). It does **not** claim to be the first multilingual or first Indic-language hallucination detector — those claims belong to Mu-SHROOM and SHROOM-CAP respectively, and must be cited and credited as such in the paper and patent filing. It also does **not** claim Marathi coverage in this project cycle; that is explicitly future work (Section 15).

Practical implication for the team: the annotator-agreement findings from Mu-SHROOM (experts struggled to

agree even on span boundaries) suggest running a small pilot annotation round (50–100 claims) before committing to the target Cohen's κ ≥ 0.70, to calibrate expectations early.

Additional closest prior art (added after further search, see Section 17 Category E): MultiHal (Lavrinovics et al., 2025) evaluates semantic similarity, NLI, and hallucination detection improvements from knowledge-graph grounding — but as separate evaluation metrics over 5 European languages (no Hindi), not as one fused claim-level score, and not as a deployable system. FactKG (Kim et al., 2023) and KGR (Guan et al., 2024) perform KG-based verification and claim-extraction-plus-verification respectively, but neither fuses NLI+semantic+KG into a single score, and neither covers Hindi. SatyaScore remains distinct from all three by combining claim-level decomposition, a single fused three-signal score, a real-time API, and Hindi coverage — together, in one system.

4. Project Overview

4.1 Project Identity

- **Project Name:** SatyaScore: Bilingual Hallucination Detection & Scoring Framework
- **Type:** AI/NLP Research System with API and Web Demo
- **Domain:** Natural Language Processing / LLM Safety / Trustworthy AI
- **Novel Contribution:** Composite Multilingual Hallucination Scoring Algorithm (CMHSA) — first fusion of semantic similarity + NLI + knowledge-graph verification at claim level, for Hindi and English
- **Target Languages:** English, Hindi (**core, committed scope**); Marathi (**tiered stretch goal — smaller dataset, lower priority, see Section 15**)
- **Marathi status (v2.3):** Reinstated as a scoped-down stretch deliverable — approximately 800–1,200 annotated claims (versus 3,000 for English/Hindi), explicitly ranked *below* the core English/Hindi goals and the ablation study in priority. If Weeks 8–11 fall behind schedule, Marathi work is the first thing cut back further toward pure future-work status, without renegotiation. The architecture (XLM-R/MuRIL embeddings, CMHSA formula) already supports Marathi natively, which is what makes this tiered approach realistic rather than aspirational.
- **Tamil status:** Out of scope. Noted only as a longer-term extension possibility beyond this capstone.
- **Team Size:** 4 members
- **Duration:** 4 months (~17 weeks) — revised from original 6-month plan
- **Publication Target:** IEEE Access, or a student/workshop track affiliated with ACL/EMNLP/NAACL
- **Patent Target:** Indian Provisional Patent — Indian Patent Office (Form 2)

4.2 What the System Does

Given any LLM-generated text in English or Hindi (Marathi planned as future extension), the system: 1. Extracts atomic factual claims from the text (Claim Extractor module). 2. Embeds each claim using cross-lingual multilingual embeddings (XLM-R primary, MuRIL fallback for Hindi). 3. Retrieves top-k relevant knowledge passages from a multilingual FAISS vector store, plus Wikidata SPARQL triples. 4. Computes a composite hallucination score per claim using NLI + semantic similarity + KG grounding (CMHSA). 5. Returns a structured JSON report with per-claim verdicts: Verified / Uncertain / Hallucinated. 6. Visualises results on an interactive Streamlit dashboard with claim-level highlighting.

4.3 Dual Purpose

Research Paper	Patent Application
New hallucination benchmark dataset in Hindi + English with 3,000+ claim-level annotations, combining semantic + NLI + KG labels not present together in any existing dataset	Novel Composite Multilingual Hallucination Scoring Algorithm (CMHSA) fusing three verification signals, explicitly including knowledge-graph grounding
Evaluation against 5 baselines (SelfCheckGPT, FActScore-style, TRUE-style NLI scorer, a FactKG-style KG-only verifier, and a MultiHal-style separate-metrics reporter) showing statistically significant improvement ($p < 0.05$)	Real-time claim-level scoring pipeline with cross-lingual knowledge-graph retrieval
Ablation study isolating the individual contribution of semantic, NLI, and KG components — proving KG grounding adds value beyond semantic+NLI alone	Multilingual extensibility framework, demonstrated concretely via the tiered Marathi dataset rather than only claimed in theory

5. Goals & Success Metrics

5.1 Primary Goals

- Achieve F1 score ≥ 0.78 (English), ≥ 0.72 (Hindi) on claim-level hallucination detection.
- Create and publicly release a benchmark dataset of ≥ 3,000 annotated claim-level examples across English and Hindi.
- **(Tiered stretch goal, ranked below the above)** Build a smaller Marathi dataset of ≥ 800 annotated claims (target 1,200) and report CMHSA performance on it — explicitly a secondary result, not equal-weight with English/Hindi.
- Demonstrate statistically significant improvement over at least 3 of 5 baselines ($p < 0.05$) — see updated baseline list in Section 4.3.
- Run an ablation study proving the knowledge-graph component contributes measurable improvement over semantic+NLI alone — this is the single most important experiment for both the paper and the patent, since it is the crux of the novelty claim.
- Prepare a Provisional Indian Patent draft by Week 14.
- Submit a full research paper draft for faculty review by Week 14, with external submission to follow shortly after.

5.2 Success Metrics Table

Metric	Minimum Target	Stretch Target
Hallucination Detection F1 (English)	0.78	0.85
Hallucination Detection F1 (Hindi)	0.72	0.80
Hallucination Detection F1 (Marathi, stretch)	0.60	0.70
Dataset Size — English + Hindi (annotated claims)	3,000	5,000
Dataset Size — Marathi (stretch, annotated claims)	800	1,200
Inter-Annotator Agreement (Cohen's κ)	0.65 (revised down from 0.70, based on Mu-SHROOM's documented annotator disagreement even at span level)	0.75
API Response Latency (per claim)	< 500ms	< 200ms
Baseline Models Outperformed	3 of 5	5 of 5
KG-Ablation Improvement (F1 delta vs. semantic+NLI only)	≥ +0.03	≥ +0.06
Paper Submission (draft ready for faculty review)	Week 14	Week 12

6. Stakeholders & Team Roles

6.1 Team Structure

Member	Role Title	Primary Responsibilities	Secondary
Member 1	ML / Model Lead	Cross-lingual embeddings (XLM-R/MuRIL), CMHSA scoring algorithm and weight-learning, model training & the 5-baseline comparison	Experiment tracking (W&B)
Member 2	Data & Annotation Lead	Dataset collection, English/Hindi annotation pipeline management, inter-annotator agreement (IAA), coordinating the Marathi stretch-dataset annotator	Wikidata SPARQL connector, KG_conf coverage testing
Member 3	Backend / Systems Engineer	FastAPI endpoint, FAISS retrieval, ablation-mode API toggle, deployment (Docker), GitHub CI/CD	Streamlit dashboard build (handed off from Member 4, see note below)
Member 4	Research & Documentation Lead	Literature review (all 5 categories	Presentation & poster design

		incl. MultiHal/FactKG/KGR differentiation), paper writing, patent draft, synopsis/report writing	

> **v2.3 rebalancing note:** In v2.2, Member 4 carried both the writing workload (literature review, paper, patent) *and* the Streamlit dashboard build — two quite different skill sets under time pressure. The dashboard has been moved to Member 3 (Backend/Systems), who is already building the API it depends on, so Member 4 can focus fully on the writing-heavy deliverables that carry real risk if rushed (patent claim precision, prior-art differentiation).

6.2 External Stakeholders

- **Faculty Mentor (Dr. S. R. Gupta)** — Technical guidance, paper co-authorship, patent co-filing support.
- **Industry Mentor (if applicable)** — Real-world use case validation, deployment feedback.
- **Annotators** — 2–3 domain experts fluent in Hindi (core) for dataset labelling and inter-annotator agreement, plus at least 1 Marathi-fluent annotator recruited within the first 2–3 weeks specifically to support the tiered Marathi stretch dataset. Recruiting this Marathi annotator early is a Phase 1 priority, not a later add-on.
- **Target Users** — Developers, researchers, and organisations deploying LLMs in critical domains.

6.3 Weekly Cadence

- **Every Monday** — Team sync (30 min): blockers, task status, next steps.
- **Every Friday** — Progress log commit to shared Notion workspace.
- **Monthly** — Full demo to faculty mentor with a written progress report.

7. System Architecture

7.1 High-Level Pipeline

Stage	Module	What It Does	Output
①	Claim Extractor	Parses LLM-generated text using dependency parsing and coreference resolution to isolate atomic verifiable claims	List of atomic claims
②	Cross-lingual Embedder	Encodes each claim using XLM-R / MuRIL to obtain language-agnostic dense vector representations	Claim embedding vectors
③	Knowledge Retriever	Performs ANN search over multilingual FAISS index (Wikipedia + Wikidata) to fetch top-k supporting passages	Top-k evidence passages
④	CMHSA Scorer	Computes composite score: HalluScore = α·SemSim + β·NLI + γ·KG_conf using trained, per-language weights	Per-claim HalluScore (0–1)
⑤	Output & Visualiser	Structures results into JSON report and renders interactive claim-level dashboard with colour-coded verdicts	Dashboard + API JSON

7.2 The CMHSA Scoring Formula (Patent Core)

HalluScore(c) = α · SemSim(c, E) + β · NLI_prob(c, E) + γ · KG_conf(c, G)

Where: - **SemSim(c, E)** — Cosine similarity between claim embedding and top-k evidence embeddings (0–1). - **NLI_prob(c, E)** — Entailment probability from a fine-tuned DeBERTa-v3 NLI model, extended to Hindi via IndicXNLI fine-tuning (0–1). - **KG_conf(c, G)** — Knowledge-graph confidence from Wikidata SPARQL triple verification (0–1). Wikidata's Hindi-language entity and label coverage is narrower than English; the team should budget time to test fallback behaviour (e.g., matching against English-labelled entities when a Hindi label is missing) and document this limitation explicitly. (This same gap is expected to be more severe for Marathi, reinforcing the decision to treat it as future work.) - **α, β, γ** — Language-specific trained weights, learned via a

softmax-constrained linear model (parameterized so α + β + γ = 1 by construction) fit on annotated training data by minimizing cross-entropy against gold verdict labels — not plain unconstrained logistic regression, which does not naturally enforce this constraint. - **Threshold:** Score < 0.35 = Hallucinated | 0.35–0.65 = Uncertain | > 0.65 = Verified.

7.3 Architecture Diagram (Textual)

```
[User Input: LLM Text in EN/HI/MR]
        |
        v
[① Claim Extractor] --> Atomic Claims List
        |
        v
[② Cross-lingual Embedder: XLM-R / MuRIL]
        |
   +----+----+
   v         v
[FAISS Index]  [Wikidata SPARQL]
(Wikipedia)     (KG Triples)
   +----+----+
        v
[④ CMHSA Scorer: α·Sem + β·NLI + γ·KG]
        |
        v
[⑤ Output: JSON + Streamlit Dashboard]
✅ Verified | ⚠ Uncertain | ❌ Hallucinated
```

8. Core Functional Requirements

FR-01: Claim Extraction Module - SHALL accept plain text input in English or Hindi (up to 2,000 words). - SHALL use dependency parsing (spaCy + IndicNLP) to extract atomic factual claims. - SHALL resolve coreferences before extraction (e.g., replace pronouns with entity names). - SHALL output a structured list of claims with character-level span offsets in the original text. - SHALL handle a minimum of 50 claims per document without performance degradation.

FR-02: Cross-Lingual Embedder - SHALL embed claims using XLM-R-large as the primary model (768-dim vectors). - SHALL use MuRIL as a fallback model for Hindi-specific inputs, particularly transliterated (Latin-script) text. - SHALL support batch embedding of up to 128 claims per forward pass. - SHALL normalise embeddings to unit vectors before similarity computation.

FR-03: Multilingual Knowledge Retriever - SHALL maintain a FAISS IVF index of multilingual Wikipedia passages, sized appropriately per language rather than a single blanket target — English Wikipedia alone can supply several million passages, but Hindi and Marathi Wikipedia are considerably smaller, so the realistic combined target is expected to be lower (to be confirmed during Phase 1 corpus profiling) rather than assumed at 5 million. - SHALL retrieve top-10 candidate passages per claim within 100ms. - SHALL query the Wikidata SPARQL API for entity-linked triple verification, with an explicit fallback to English-labelled entities when a Hindi label is unavailable. - SHALL support index updates without a full rebuild (incremental indexing). - SHALL cache SPARQL query results with a 24-hour TTL to reduce API calls.

FR-04: CMHSA Scoring Engine (Core) - SHALL compute three sub-scores per claim: SemSim, NLI_prob, KG_conf. - SHALL apply language-specific weights (α, β, γ) learned during training. - SHALL output a final HalluScore in [0.0, 1.0] with a confidence interval. - SHALL classify claims into three verdict tiers: Verified (>0.65), Uncertain (0.35–0.65), Hallucinated (<0.35). - SHALL support custom threshold configuration via API parameters. - SHALL support an ablation mode (toggle individual signals on/off) to reproduce the paper's ablation study directly from the API.

FR-05: REST API (FastAPI) - SHALL expose a POST /detect endpoint accepting JSON with fields: text (str), language (str), threshold (float, optional). - SHALL return JSON with: claims (list), scores (list), verdicts (list), overall_score (float), processing_time_ms (int). - SHALL include API key authentication and rate limiting (100 req/min per key). - SHALL provide Swagger/OpenAPI documentation at /docs. - SHALL return HTTP 422 with a

descriptive error for invalid inputs.

FR-06: Web Demo Dashboard - SHALL provide a Streamlit-based web interface accessible at localhost:8501. - SHALL highlight hallucinated claims in RED, uncertain in AMBER, verified in GREEN within the original text. - SHALL display a bar chart of HalluScore distributions across all claims. - SHALL allow users to export results as CSV or JSON. - SHALL support file upload (.txt, .pdf) in addition to text pasting.

FR-07: Benchmark Dataset - SHALL contain ≥ 3,000 claim-level examples across English and Hindi. - SHALL be balanced: ~40% Hallucinated, ~30% Uncertain, ~30% Verified. - SHALL include inter-annotator agreement scores (target κ ≥ 0.65, stretch ≥ 0.75). - SHALL be released publicly on HuggingFace Datasets under CC-BY-4.0 license, with an explicit data card describing the combined semantic+NLI+KG annotation scheme.

9. Non-Functional Requirements

Category	Requirement	Target
Performance	End-to-end latency per document (10 claims)	< 3 seconds on GPU
Performance	FAISS retrieval latency per claim	< 100ms
Scalability	Concurrent API requests supported	≥ 20 simultaneous
Accuracy	F1 score on English benchmark	≥ 0.78
Accuracy	F1 score on Hindi benchmark	≥ 0.72
Reliability	API uptime during demo period	≥ 99%
Reproducibility	All experiments reproducible with fixed random seeds	100% deterministic
Documentation	GitHub README with setup, usage, and API docs	Complete
Code Quality	Test coverage for core modules	≥ 70%
Portability	Docker container for one-command deployment	Required

10. Technology Stack (with Concrete Hardware Specs)

Layer	Component	Technology / Library	Purpose
Language	Runtime	Python 3.10+	Primary development language
ML Framework	Deep Learning	PyTorch 2.0	Model training & inference
NLP Models	Cross-lingual Embedder	XLM-R-large (HuggingFace)	Multilingual claim embedding
NLP Models	Indian Language Model	MuRIL (Google)	Hindi specialisation
NLP Models	NLI Scorer	DeBERTa-v3-large fine-tuned on MNLI + IndicXNLI	Entailment probability
NLP Models	Claim Extractor	spaCy 3.x + IndicNLP	Dependency parsing, NER
Retrieval	Vector Store	FAISS (IVF flat index)	ANN similarity search
Knowledge Graph	Fact Verification	Wikidata SPARQL API	Entity triple lookup
Backend	API Server	FastAPI + Uvicorn	REST API & serving
Frontend	Demo Dashboard	Streamlit	Interactive web UI
Experiment Tracking	Logging	Weights & Biases (W&B)	Metrics, curves, artefacts
Data Versioning	Dataset Management	DVC + Git LFS	Reproducible data pipelines
Containerisation	Deployment	Docker + Docker Compose	One-command setup
Version Control	Code	GitHub (private → public post-patent)	Source control & CI

10.1 Concrete Hardware Requirements (New in v2.0)

For training / fine-tuning (NLI model, weight learning): - GPU: minimum 1× NVIDIA T4 (16GB VRAM) — available via Google Colab Pro+; recommended 1× NVIDIA A100 (40GB) or V100 (16–32GB) via AWS EC2 g4dn.xlarge/p3.2xlarge for faster fine-tuning of DeBERTa-v3-large. - RAM: ≥ 32GB system RAM recommended for FAISS index construction over 5M+ passages. - Storage: ≥ 100GB free disk space (Wikipedia dump + embeddings + FAISS index + model checkpoints; the FAISS IVF index alone over 5M 768-dim vectors is roughly 15–20GB).

For inference / API serving (demo/deployment): - CPU-only inference is feasible for the API layer (FastAPI +

FAISS search) at reduced throughput; GPU recommended for real-time NLI scoring to meet the <500ms latency target. - Minimum: 1× NVIDIA T4 or equivalent, 16GB RAM, 4 vCPUs for the demo deployment.

For annotation team: - Standard laptops/desktops sufficient; annotation interface should be a lightweight web tool (e.g., Label Studio) requiring no GPU.

11. Dataset Strategy

11.1 Existing Datasets (Baseline & Training)

Dataset	Language	Size	Use in This Project
TruthfulQA	English	817 questions	Baseline benchmark evaluation
HaluEval	English	10,000 examples	Hallucination classifier pre-training
FActScore Benchmark	English	~500 biographies	Fine-grained scoring comparison
IndicQA	Hindi, Bengali	~6,000 QA pairs	Indian language factual queries
IndicXNLI	Hindi + 10 Indian languages	Translated XNLI scale	Fine-tuning NLI_prob component for Hindi
Mu-SHROOM (SemEval-2025)	14 languages incl. Hindi	Shared-task scale	Cross-comparison baseline; NOT used as sole novelty claim
Multilingual Wikipedia	100+ languages	Billions of tokens	FAISS knowledge base
FEVER	English	185,445 claims	Fact verification pre-training
XQuAD	Hindi + 10 others	1,190 paragraphs	Cross-lingual QA evaluation
MahaSQuAD / IndicSQuAD	Marathi	SQuAD-scale	Accelerates Marathi query generation for the tiered stretch dataset (Section 15.1)
L3Cube-MahaCorpus	Marathi	~24.8M sentences	Seeds the Marathi portion of the FAISS index

11.2 Your Original Dataset (Research Contribution)

Creating this dataset is the most important research contribution. Without it, the claim-level, KG-grounded novelty claim is unsubstantiated.

- **Step 1 — Query Generation:** Create 1,500+ factual questions in English and Hindi covering Healthcare, History, Science, Government, and Sports domains. **(Tiered stretch, Phase 4):** an additional 400-600 Marathi questions, substantially accelerated by drawing from MahaSQuAD/IndicSQuAD rather than writing from scratch.
- **Step 2 — LLM Response Generation:** Feed queries to GPT-4, Llama 3, and Gemini. Each generates a multi-sentence response = potential hallucinations.
- **Step 3 — Claim Decomposition:** Use the Claim Extractor module to decompose responses into atomic claims (~3,000+ claims total).
- **Step 4 — Expert Annotation:** 2–3 annotators label each claim as Verified / Uncertain / Hallucinated, with an evidence URL.
- **Step 5 — Pilot Round (New in v2.0):** Before full-scale annotation, run a 50–100 claim pilot batch specifically to test annotator agreement and refine guidelines — informed directly by Mu-SHROOM's documented annotator disagreement on span boundaries.
- **Step 6 — IAA Calculation:** Compute Cohen's Kappa between annotator pairs. Resolve disagreements by majority vote or adjudication.
- **Step 7 — Dataset Release:** Upload to HuggingFace Datasets under CC-BY-4.0, with a data card and paper citation, explicitly noting the combined semantic+NLI+KG claim-level annotation scheme as the dataset's key distinguishing feature.

11.3 Annotation Guidelines Summary

- **Verified:** Claim is factually accurate and confirmed by ≥ 2 independent sources.
- **Hallucinated:** Claim contains specific false or fabricated information contradicted by sources.
- **Uncertain:** Claim cannot be confirmed or denied due to ambiguity, opinion, or lack of sources.

12. Development Roadmap (4 Months / ~17 Weeks)

> **Timeline (v2.3):** Confirmed as a full 4 months (~17 weeks). Critically, **annotation now runs in parallel with pipeline development starting in Phase 2**, rather than being crammed entirely into one later phase — this was flagged as the most fragile part of the v2.2 roadmap during strict review, since stacking full annotation + baselines + ablation into a single 4-week phase was unrealistic. Marathi's tiered stretch dataset is explicitly sequenced *after* the English/Hindi dataset is on track, so it never competes with the core deliverables for time.

Phase	Weeks	Key Deliverables	Owner
Phase 1 — Foundation	Weeks 1–3	Literature review finalized (incl. MultiHal/FactKG/KGR differentiation), GitHub repo set up, baseline cosine-similarity system running, claim extractor (English) built, dev environment configured, **Marathi-fluent annotator recruited**	All Members
Phase 2 — Core Build + Annotation Start	Weeks 4–8	XLM-R + MuRIL embedder integrated, FAISS index built (sized per Phase 1 corpus profiling), NLI scorer connected (DeBERTa-v3 + IndicXNLI fine-tune), CMHSA v1 formula implemented with softmax-constrained weight learning, pilot annotation (50–100 claims) run and guidelines refined, **full-scale English/Hindi annotation begins in parallel**	M1 + M2 + M3
Phase 3 — Dataset Completion & Experiments	Weeks 9–12	English/Hindi annotation completed (3,000 claims), end-to-end pipeline complete, 5-baseline comparison implemented (prioritize FactKG-style and TRUE-style first if time-constrained), ablation study run (semantic vs. semantic+NLI vs. full CMHSA — non-negotiable core experiment), IAA calculated	M1 + M2
Phase 4 — Marathi Stretch Tier	Weeks 12–13 (only if Phase 3 is on schedule)	Marathi dataset annotation (800–1,200 claims) using guidelines refined from the Hindi pilot, CMHSA run on Marathi subset, results reported as a clearly-labeled secondary/stretch result	M2 (Marathi annotator) + M1
Phase 5 — Paper Writing & Patent Draft	Weeks 13–15	Full paper draft, related-work section differentiating from Mu-SHROOM/SHROOM-CAP/MultiHal/FactKG/KGR, provisional patent draft prepared, faculty mentor review	M4 + All
Phase 6 — Final Polish & Submission	Weeks 16–17	API hardened, dashboard polish (Member 3), Docker deployment, paper submitted, dataset(s) released on HuggingFace, final report and viva/demo preparation	All Members

If time runs short, drop in this order: (1) Marathi stretch tier entirely — reverts cleanly to pure future-work status, no core deliverable lost, (2) stretch dataset size beyond 3,000 for English/Hindi, (3) 5th baseline comparison (keep at minimum 3), (4) Docker containerization polish, (5) dashboard CSV/PDF export niceties. **Never drop:** the ablation study — it is the single experiment your entire novelty claim depends on.

13. Research Paper Strategy

- **Target venue:** IEEE Access (open access, faster turnaround, receptive to applied NLP systems) or a workshop track affiliated with ACL/EMNLP/NAACL suited to student capstone submissions.
- **Related-work framing:** Must explicitly position against Mu-SHROOM, SHROOM-CAP, MultiHal, FactKG, and KGR as the closest prior art – crediting their contributions while clearly stating the paper's defensible novelty (claim-level, single-fused semantic+NLI+KG score, with a real-time API, for Hindi as the core language and Marathi as a demonstrated tiered extension).
- **Core experiment:** The ablation study (semantic-only vs. semantic+NLI vs. full CMHSA) is the paper's central empirical claim and should be prioritized over any other experiment if time runs short.

14. Patent Strategy

- **Target:** Indian Provisional Patent Application (Form 2), Indian Patent Office.
- **Patentable core:** The specific fusion mechanism of CMHSA (the weighted combination of semantic similarity, NLI entailment, and knowledge-graph confidence into a single claim-level score, with language-specific learned weights) — not the individual components (XLM-R, MuRIL, DeBERTa, Wikidata are all pre-existing, non-patentable building blocks).
- **Recommendation:** Keep the exact weight-learning methodology and threshold-calibration approach as the confidential "specific implementation" detail in the provisional filing, while the research paper can describe the general architecture more openly.

15. Risk Assessment & Mitigation

Risk	Likelihood	Impact	Mitigation
Wikidata's Hindi entity/label coverage is sparser than English, reducing reliable KG_conf scores	Medium	Medium-High (affects core novelty signal)	Test fallback-to-English-label matching early (Month 2); document coverage gaps transparently as a limitation rather than hiding them
Inter-annotator agreement falls below target κ	Medium	Medium	Run pilot annotation round first (Month 2); refine guidelines before full-scale annotation
3,000-claim dataset proves too small for robust results	Low-Medium	Medium	Follow SHROOM-CAP's precedent of combining multiple existing datasets if native data proves insufficient
Team underestimates compute needs for DeBERTa-v3-large fine-tuning	Medium	Medium	Reserve AWS/Colab Pro+ credits early; have a smaller-model (DeBERTa-v3-base) fallback plan
Novelty claim challenged by reviewers/patent examiner due to Mu-SHROOM/SHROOM-CAP overlap	Medium	High	Explicit, honest positioning (Section 3) prepared in advance; ablation study as primary evidence of genuine contribution
4-month timeline slippage	Medium	Medium	Bi-weekly faculty check-ins; prioritize the ablation study and the English/Hindi dataset over stretch goals (Marathi tier, Tamil, 5,000-claim dataset) if behind schedule
Ablation study shows KG grounding does not meaningfully improve results over semantic+NLI alone	Medium	Critical (undermines core novelty claim)	Pre-register this as a possible outcome rather than assuming success; if it occurs, report it honestly as a scientific finding (a well-documented negative result is still publishable) and shift patent claim emphasis toward the fusion methodology and real-time claim-level architecture itself, rather than

			KG-specific performance gains
Marathi annotator not recruited in time (Phase 1)	Medium	Low (affects only the stretch tier, not core deliverables)	Marathi tier is explicitly the first thing dropped if this slips — core English/Hindi deliverables are unaffected by design
AWS/GPU compute costs exceed available budget	Low-Medium	Medium	Confirm available compute budget (department funding, personal, or free-tier Colab Pro+) during Phase 1; DeBERTa-v3-base fallback reduces cost if needed

15.1 Marathi: Tiered Stretch-Goal Plan (Revised in v2.3)

Marathi is now a **committed but explicitly lower-priority tier**, not a core deliverable and not pure future-work — the distinction matters: unlike v2.1/v2.2, the team will attempt real annotation and evaluation this cycle, sequenced so it never displaces the English/Hindi core goals or the ablation study.

- **Reusable components:** XLM-R already covers Marathi out of the box; MuRIL was explicitly pre-trained on Marathi among its 17 languages, so the embedding layer needs no architectural change.
- **Reusable text/QA resources:** L3Cube-MahaCorpus (≈24.8M Marathi sentences) can seed the Marathi portion of the FAISS index; MahaSQuAD and IndicSQuAD provide existing Marathi factual QA pairs that can accelerate query generation instead of writing 500+ Marathi questions from scratch; IndicXNLI already includes Marathi entailment data for NLI_prob fine-tuning.
- **Scope for this cycle:** 800–1,200 annotated claims (vs. 3,000 for English/Hindi), using annotation guidelines already refined from the Hindi pilot round (Phase 2) rather than developing separate guidelines from zero. One dedicated Marathi-fluent annotator, recruited in Phase 1.
- **What happens if time runs short:** Per Section 12's drop order, the Marathi tier is the *first* thing cut back — reverting cleanly to pure future-work status (as described in earlier PRD versions) with zero impact on core deliverables. This is a deliberate design choice, not a risk being hidden.
- **Recommended framing for the synopsis and paper:** Present Marathi results as a clearly-labeled secondary/stretch finding — e.g., "a smaller-scale Marathi evaluation, included as a lower-priority extension" — rather than claiming full parity with the English/Hindi benchmark. This is both intellectually honest and strengthens the "extensibility" argument in the patent filing (Section 14), since it demonstrates the architecture generalizing in practice, not just in theory.

16. Ethical & Responsible-AI Considerations

- **Domain sensitivity:** Since target use cases include healthcare and government-advisory content, the team should avoid using real, potentially sensitive medical or legal queries during dataset construction without appropriate care; synthetic/generic factual queries are preferred over queries mimicking real patient or citizen scenarios.
- **Dataset bias:** Annotators' own linguistic and regional backgrounds may influence what counts as "common knowledge" in Hindi-language contexts — guidelines should be reviewed for cultural neutrality.
- **Transparency in limitations:** Any published paper or patent filing should transparently state the KG coverage gap between English and Hindi (and the likely larger gap for Marathi), and should clearly label Marathi results as a smaller-scale, lower-priority tier rather than implying full parity with the English/Hindi benchmark.
- **Dual-use consideration:** A hallucination detector could in principle be used to identify weaknesses in a competitor's LLM for adversarial purposes; the team should frame the tool's intended use as defensive/auditing, consistent with its healthcare/governance motivation.

17. Appendix: Key Reference Papers (Updated — Now 27 Papers Across 5 Categories)

A. Foundational Hallucination Detection: FEVER (Thorne et al., 2018), FActScore (Min et al., 2023), SelfCheckGPT (Manakul et al., 2023), TruthfulQA (Lin et al., 2022), HaluEval (Li et al., 2023), TRUE (Honovich et al., 2022), Huang et al. survey (2024/25), Hu et al. Knowledge-Centric Detection (2024), ANAH (Ji et al., 2024), Mishra et al. Fine-grained Editing (2024), Su et al. Internal-States Detection (2024).

B. Multilingual Hallucination Detection (closest language-coverage prior art): Mu-SHROOM / SemEval-2025 Task 3 (Vázquez et al., 2025) and system papers (MSA, HausaNLP, Hallucination Detectives); SHROOM-CAP (2025).

C. Cross-lingual Embeddings & Indic Models: XLM-R (Conneau et al., 2020), MuRIL (Khanuja et al., 2021), IndicBERT (Kakwani et al., 2020), IndicSBERT (Deode et al., 2023), IndicXNLI (Aggarwal et al., 2022).

D. Fact Verification / RAG: FEVER (Thorne et al., 2018), Retrieval-Augmented Generation (Lewis et al., 2020), TRUE (Honovich et al., 2022).

E. Knowledge-Graph-Grounded Hallucination Detection (NEW — most critical category, tests the core novelty claim directly): - **MultiHal** (Lavrinovics et al., 2025) — *MultiHal: Multilingual Dataset for Knowledge-Graph Grounded Evaluation of LLM Hallucinations*, arXiv 2505.14101. **Closest prior art identified to date.** Multilingual (Spanish, French, Italian, Portuguese, German — no Hindi), Wikidata-grounded, reports semantic similarity + NLI + hallucination-detection improvements from KG-RAG as separate evaluation metrics. Critical distinction: MultiHal is an **evaluation benchmark measuring these signals separately**; SatyaScore **fuses them into one deployable, claim-level confidence score with a real-time API**, and covers Hindi, which MultiHal does not. - **Agrawal et al. (2024)** — *Can Knowledge Graphs Reduce Hallucinations in LLMs?: A Survey*, NAACL 2024. Direct survey on this exact research question. - **Kim et al. (2023)** — *FactKG: Fact Verification via Reasoning on Knowledge Graphs*, ACL 2023. Close architectural cousin (KG-based reasoning for verification); no fusion with NLI/semantic signals, no Hindi coverage. - **Guan et al. (2024)** — *KGR* — hallucination detection via dedicated claim extraction, fact selection, and verification modules. Structurally close to SatyaScore's pipeline shape; requires explicit differentiation in the Literature Survey (see note below), not just a citation. - **Lavrinovics et al. (2025)** — *Knowledge Graphs, Large Language Models, and Hallucinations: An NLP Perspective*, Journal of Web Semantics. Broader survey, useful for framing the Literature Survey's opening context on KG+LLM hallucination research.

> **Note for Literature Survey drafting:** Category E papers must be addressed head-on, not folded in passively. MultiHal, FactKG, and KGR are collectively the closest work to SatyaScore found in this review. The differentiation to state explicitly, each time one of these is cited: SatyaScore is the first to combine claim-level decomposition + a single fused (not separately-reported) semantic+NLI+KG score + a real-time deployable API + Hindi-language coverage, in one general-domain system. No reviewed paper — including this new category — combines all four of these together.

End of PRD v2.3. This document is the authoritative project reference going forward and should be updated as findings evolve — particularly after the Month 2 Wikidata-Hindi-coverage test and the Month 2 pilot annotation round, both of which may require further scope adjustments. Marathi extension should be revisited only after core English/Hindi goals (Section 5) are secured. Category E (KG-grounded hallucination detection) was added after discovering MultiHal, FactKG, and KGR as closer prior art than previously known — the novelty claim in Section 3 remains valid but must now explicitly differentiate from these three works wherever it is stated, in the synopsis, paper, and patent filing alike.